北京郵電大學

Queen Mary University of London

Joint Programme Assessments 2022/23

Answer Sheet

(To be used in conjunction with the provided alternative assessment questions and the supplementary instructions)

Complete the information below about yourself very carefully.

QM student number	200978947
BUPT student number	2020213248
Class number	2020215107
Module code	EBU5302
Word Count	

I hereby confirm that the submitted work is entirely my own and I have not (i) used the services of any agency or person(s) providing specimen, model or ghostwritten work in the preparation of the work I submit for this open book assessment; (ii) given assistance in accessing this paper or in providing specimen, model or ghostwritten work to other candidates submitting for this open-book assessment.

Signature *Tangxun Wu*

For examiners ONLY

Do not write in this section

	Q1	Q2	Q3	Q4			Sum
A							
B							
C							
D							
E							
F							
TOTAL							

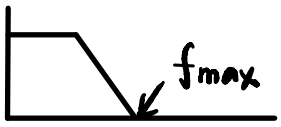
Question	1	Do not write in this column
Section(s)	a. b.	

Q1

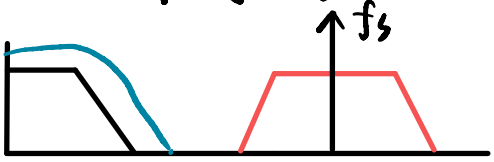
a) ∵ Bandwidth $B = 35\,kHz$

∴ $f_{s\,min} = 2B = 70\,kHz$

b) original signal

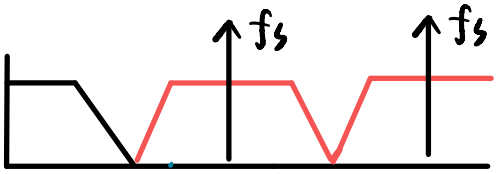


oversampling: signal sampled with $f_s > 2f_{max}$:



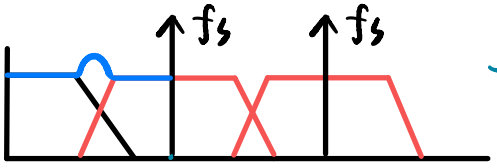
the low-pass filter

sampled with $f_s = 2f_{max}$:



need an ideal filter to get the original signal

sampled with $f_s < 2f_{max}$:



spectra interfere → can't recover original signal, have aliasing

Therefore, from the diagram, we know that:
oversampling can produce a relatively big gap for the original data to be filtered and extracted
So filter doesn't need to be ideal.
It's easier to design a simple filter.

Question	1	Do not write in this column
Section(s)	c. d. e.	

c)

$I = \sum \log_2(\frac{1}{p})$

$= \log_2(\frac{1}{0.17}) + \log_2(\frac{1}{0.49}) + \log_2(\frac{1}{0.25}) + \log_2(\frac{1}{0.09})$

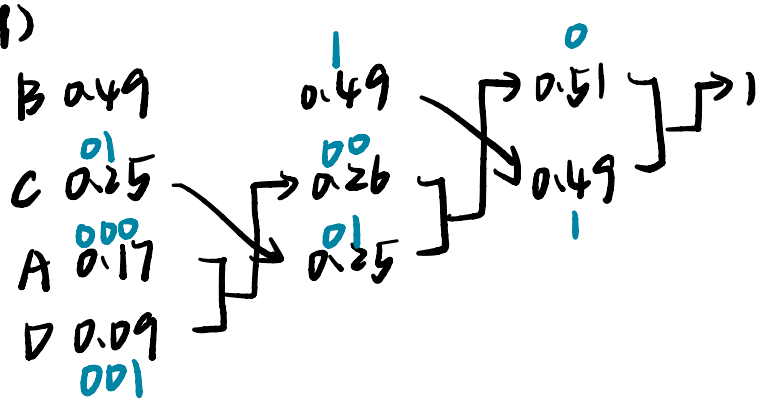
$= 9.0595$ → the information content

$H = \sum_i p_i \log_2(\frac{1}{p_i})$

$= 0.17 \log_2(\frac{1}{0.17}) + 0.49 \log_2(\frac{1}{0.49}) + 0.25 \log_2(\frac{1}{0.25}) + 0.09 \log_2(\frac{1}{0.09})$

$= 1.7515$ → the source entropy

d)



∴

symbol	A	B	C	D
code	000	1	01	001
length	3	1	2	3

∴ the average code length $= \sum p_i n_i$

$= 3 \times 0.17 + 1 \times 0.49 + 2 \times 0.25 + 3 \times 0.09 = 1.77$

e)

max entropy: $H_{max} = \log_2 N = \log_2 4 = 2$

Source efficiency: $\eta_{source} = \frac{H}{H_{max}} = \frac{1.7515}{2} = 87.575\%$

Question	2	Do not write in this column
Section(s)	1	

Q2

1) capacity $C = \max I(X;Y)$

$$I(X;Y) = \sum_{x,y} p(x,y) \log_2 \left(\frac{p(x,y)}{p(x)p(y)}\right)$$

∵ For X: $p(a) = p(b) = p(c) = \frac{1}{3}$

For Y: $p(a') = \frac{1}{3}\times 0.5 + \frac{1}{3}\times 0.25 + \frac{1}{3}\times 0.25 = \frac{1}{3}$

$p(b') = \frac{1}{3}\times 0.5 + \frac{1}{3}\times 0.25 + \frac{1}{3}\times 0.25 = \frac{1}{3}$

$p(c') = \frac{1}{3}\times 0.5 + \frac{1}{3}\times 0.25 + \frac{1}{3}\times 0.25 = \frac{1}{3}$

$p(a,a') = \frac{1}{6}$ $p(a,b') = \frac{1}{12}$ $p(a,c') = \frac{1}{12}$

$p(b,a') = \frac{1}{12}$ $p(b,b') = \frac{1}{6}$ $p(b,c') = \frac{1}{12}$

$p(c,a') = \frac{1}{12}$ $p(c,b') = \frac{1}{12}$ $p(c,c') = \frac{1}{6}$

$$\therefore I(X;Y) = \frac{1}{6}\log_2\frac{\frac{1}{6}}{\frac{1}{3}\times\frac{1}{3}} + \frac{1}{12}\log_2\frac{\frac{1}{12}}{\frac{1}{3}\times\frac{1}{3}} + \frac{1}{12}\log_2\frac{\frac{1}{12}}{\frac{1}{3}\times\frac{1}{3}}$$
$$+ \frac{1}{12}\log_2\frac{\frac{1}{12}}{\frac{1}{3}\times\frac{1}{3}} + \frac{1}{6}\log_2\frac{\frac{1}{6}}{\frac{1}{3}\times\frac{1}{3}} + \frac{1}{12}\log_2\frac{\frac{1}{12}}{\frac{1}{3}\times\frac{1}{3}}$$
$$+ \frac{1}{12}\log_2\frac{\frac{1}{12}}{\frac{1}{3}\times\frac{1}{3}} + \frac{1}{12}\log_2\frac{\frac{1}{12}}{\frac{1}{3}\times\frac{1}{3}} + \frac{1}{6}\log_2\frac{\frac{1}{6}}{\frac{1}{3}\times\frac{1}{3}}$$
$$= \frac{1}{2}\log_2\frac{9}{8}$$
$$= 0.0849$$

∴ Capacity = 0.0849

Question	2	Do not write in this column
Section(s)	2	

2)

From the diagram, we get:

parity check matrix H: $\left[\begin{array}{cccc|ccc} I_1 & I_2 & I_3 & I_4 & P_1 & P_2 & P_3 \\ \hline 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$

$\because H = [P^t \mid I_{n-k}]$

$\therefore P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \Rightarrow G_1 = [I_k \mid P] = \left[\begin{array}{cccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$

① For digits 1010:

codeword $c = xG_1 = [1010010]$

∴ the parity check bits: 010

② For digits 1001:

codeword $c = xG_1 = [1001001]$

∴ the parity check bits: 001

③ For digits 0101:

codeword $c = xG_1 = [0101101]$

∴ the parity check bits: 101

Question	2	Do not write in this column
Section(s)	3	

3)

$$H = \left[\begin{array}{cccc|ccc} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right], \quad H^t = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$r = 1010111$

$\therefore$ the syndrome $s = rH^t$

$= [101]$

$\because s = eH^t$ and the max bit error per code $= 1$

$\therefore$ error sequence $e = [0010000]$

Therefore, we can have transmitted code word:

$c = r + e = 1000111$

And the information sequence can be identified by the first 4 bits: $X = 1000$.

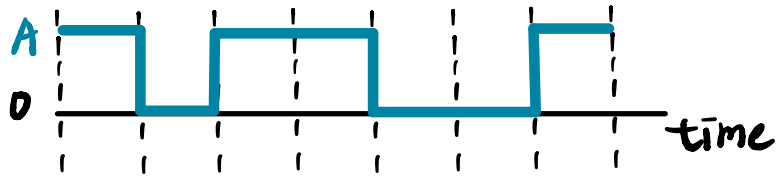
Question	3	Do not write in this column
Section(s)	a	

Q3

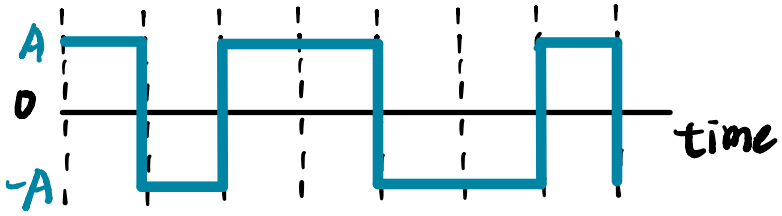
a)

1 0 1 1 0 0 1

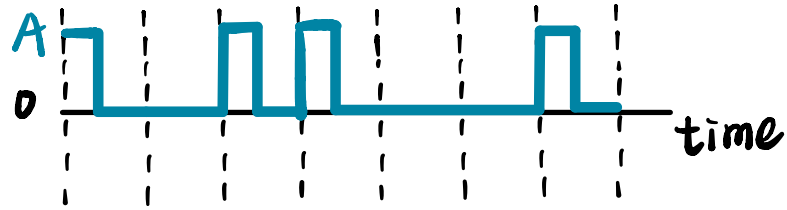
unipolar NRZ



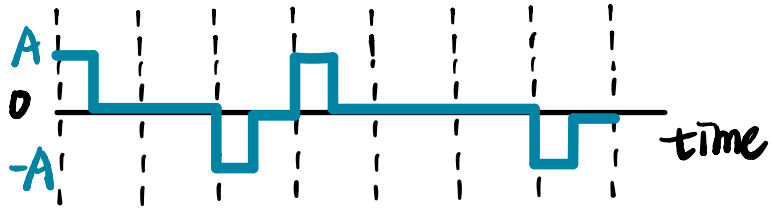
polar NRZ



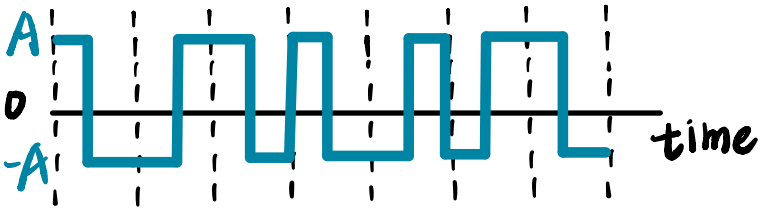
unipolar RZ



Bipolar RZ



Manchester NRZ



Question	3	Do not write in this column
Section(s)	b. c.	

b)

① the timing error allowed on the sampler at the receiver side can be given by the width inside the eye, which is called the eye opening. Besides, the place where the vertical opening of the eye is largest is the preferred time for sampling.

② The sensitivity to timing error can be given by the slope of the eye opening, and it is evaluated at, or near the 0-crossing point.

③ The noise margin of the system can be given by the height of the eye opening.

c) For polar NRZ:

∵ the possible level for a's are $+A$, $-A$ V

$R(k) = \sum_{i=1}^{L} (a_n a_{n+k}) i P_i$

∴ ① When $k=0$, $a_n a_{n+k}$ can be A^2, $(-A)^2$, each holds the probability of $\frac{1}{2}$

∴ $R(0) = \frac{1}{2}A^2 + \frac{1}{2}(-A)^2 = A^2$

② When $k \neq 0$, $a_n a_{n+k}$ can be A^2, $(-A)(+A)$, $(+A)(-A)$, $(-A)^2$, each holds the probability of $\frac{1}{4}$

∴ $R(k) = \frac{1}{4} \times A^2 + \frac{1}{4}(+A)(-A) + \frac{1}{4}(-A)(+A) + \frac{1}{4}(-A)^2$

$= 0$

$\Rightarrow R(k) = \begin{cases} A^2, & k=0 \\ 0, & k \neq 0 \end{cases}$

Question	3	Do not write in this column
Section(s)	d	

d) data rate $R = 1\,Mb/s$

$l = 4$ bit

$\therefore D = \frac{R}{l} = \frac{1}{4} \times 10^6$ symbol/second

$\therefore B_{min} = \frac{1}{2}D = \frac{1}{8} \times 10^6$ symbol/second

$= 125\,kHz$

Question	4	Do not write in this column
Section(s)	a . b	

Q4

a) as we know, diversity gain is about 4-6 dB. (up to 6 dB)

∵ frequency reuse distance $Ru = r\sqrt{3N}$

And for received power $P \propto \frac{1}{r^4}$

∴ $P = \frac{P_0}{Ru^4} = \frac{P_0}{r^4 9N^2}$

∵ For each case, there are 6 nearest co-channel neighbors

∴ $P = \frac{6P}{r^4 9N^2}$

and $\frac{P_1}{P_3} = \frac{\frac{6P}{(r\sqrt{3\times7})^4}}{\frac{6P}{(r\sqrt{3\times3})^4}} = \frac{3^2}{7^2} = \frac{9}{49} \approx -7.4\,dB$

∴ If 6 dB gain is achieved by adding the diversity -7.4 dB reduction is more or less cancelled out.

b)

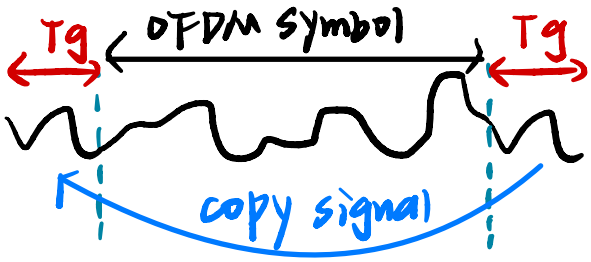
∵ $\frac{Eb}{No} = \frac{S}{KTR} = SdBW - 10\log R - 10\log k - 10\log T$

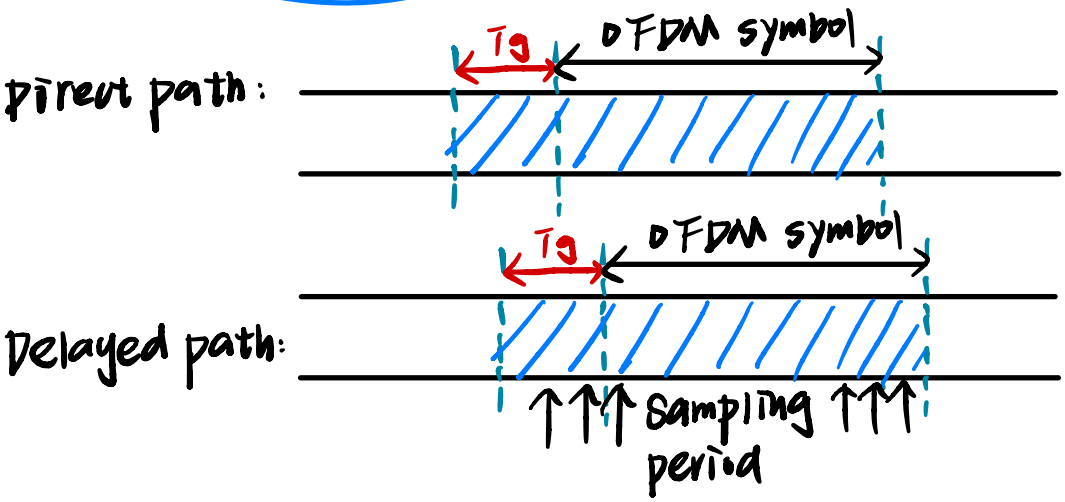
∴ $\frac{Eb}{No} = -151\,dBW - 10\log 2400 + 228.6\,dBW - 10\log 1500$

$= 12.037\,dBW$

Question	4	Do not write in this column
Section(s)	c.d.	

c) In the time domain:





∴ By adding the Guard Interval period, ISI can be avoided.

d) $\lambda = \frac{c}{f} = \frac{3\times10^8}{4\times10^9} = 0.075\ m$

free space loss $L_{dB} = 20\log_{10}\left(\frac{4\pi d}{\lambda}\right)$

$= -20\log(\lambda) + 20\log(d) + 20\log(4\pi)$

$= 195.576\ dB.$

Question		Do not write in this column
Section(s)		

Question		Do not write in this column
Section(s)		